

Assignment-1

Object-Oriented Programming (OOP)

Unit -1

1. What is Java?
2. Why is Java called a platform-independent programming language?
3. What is bytecode?
4. Differentiate between JVM, JRE, and JDK.
5. What is the role of the JIT (Just-In-Time) Compiler?
6. Explain the features of Java with suitable real-life examples.
7. Differentiate between Procedural Programming and Object-Oriented Programming.
8. Explain the four main features (pillars) of Object-Oriented Programming with examples.
9. Draw and explain the JVM Architecture.
10. Differentiate between JVM, JRE, and JDK with a neat diagram.
11. Explain the Java Compilation and Execution Process with a neat diagram.
12. Explain the structure of a Java program with an example.
13. Explain the history of Java and its evolution.
14. Explain the installation of JDK and the purpose of PATH and CLASSPATH variables.
15. What is an IDE? Explain the features and advantages of IDEs used for Java programming.

Unit -2

1. Explain the types of variables in Java with examples.
2. Differentiate between Primitive and Reference Data Types with examples.
3. Explain Type Casting in Java with examples.
4. What are Wrapper Classes? Explain Autoboxing and Unboxing with suitable examples.
5. Explain the different types of operators in Java with examples.
6. What is Operator Precedence? Explain with suitable examples.
7. Explain the looping statements in Java (for, while, do-while, and for-each) with examples.
8. Differentiate between for, while, and do-while loops with suitable examples.
9. Explain the break and continue statements with suitable examples.
10. Explain the decision-making statements in Java (if, if-else, nested if, and switch) with examples.

Unit-3

1. Explain One-Dimensional Arrays in Java with a suitable example.
2. Explain Multi-Dimensional Arrays in Java with a suitable example.
3. Explain various array operations in Java with suitable examples.
4. Explain the String class and its commonly used methods with suitable examples.
5. Differentiate between String, StringBuffer, and StringBuilder.
6. Explain String comparison methods (==, equals(), and compareTo()).
7. Explain any six String manipulation methods with suitable examples.
8. Explain String immutability and its advantages.
9. Differentiate between the == operator and the equals() method in Java.
10. Write Java programs for matrix operations using two-dimensional arrays.
